MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 2E24

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Beam power amplifier, F1 = 125 MHz

Outline --- See figure 1

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	Cap
Element	---	g3, int sd, fct	f	g2	g3, int sd, fct	g1	g3, int sd, fct	-f	Base sleeve	a

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pp
Unit:	Vdc (Note 1)	Vdc	Vdc	Vdc	mAdc	mAdc	W
Maximum:							
Class C telephony	6.9	500	-175	200	70	3.5	9
Class C telegraphy	6.9	600	-175	200	85	3.5	13.5
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	---	---	---	---	---

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Pg2	Pi	Maximum duty cycle	Modulation	Alt
Unit:	W	W	---	---	ft
Maximum:					
Class C telephony:	2.3	27	---	Anode	10,000
Class C telegraphy:	2.5	40	Note 2	Anode	10,000
Minimum:	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1236	Power oscillation (2)	Eb = 400 Vdc; Ec2 = vary, not to exceed 200 Vdc, Ec1 = -45 Vdc; Ic1 = 3 mAdc; Ib = 75 mAdc; F = 125 MHz	---	---	Po	16	---	W
Ⓐ 1216	Base material insulating quality	Low-loss phenolic Zone 11, min.	---	---	---	---	---	---
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 300 Vdc; Ec2 = 200 Vdc; Ec1 Ib = 20 mAdc (see note 3)	0.65	II	Ic1	0	-5	µAdc
1256	Electrode current (anode)	Eb = 200 Vdc; Ec2 = 135 Vdc; Ec1 = -5 Vdc	0.65	II	Ib	24	46	mAdc
1256	Electrode current (screen)	Eb = 200 Vdc; Ec2 = 135 Vdc; Ec1 = -5 Vdc	0.65	II	Ic2	---	5	mAdc
1231	Emission	Eb = Ec2 = Ec1 = 25 Vdc (see note 3)	0.65	II	Is	100	---	mAdc
Ⓐ 1201	Short and discontinuity detection	See note 5	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Eb = 250 Vdc; Ec2 = 200 Vdc; Ec1 Ib = 10 mAdc, Rp = 2,000 ohms	---	---	Ep	---	1,000	mVac
1301	Filament current		---	---	If	590	710	mAdc
1236	Power oscillation (1)	Eb = 500 Vdc; Ec2 = 200 Vdc; Ib = 60 mAdc; Ic1 = 3 mAdc; Rg1 = 15,000 ohms; F = 15 MHz	---	---	Po	18	---	W
1331	Direct-interelectrode capacitance	Without shield (see note 4)	---	---	Cgp Cin Cout	--- 7 4.9	0.11 10 8.1	pF pF pF
Ⓐ 1101	Secureness of base, cap, or insert		---	---	---	---	---	---
Ⓐ 1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
1501	Intermittent life	Group B: Eb = 500 Vdc; Ec2 = 200 Vdc; Ec1 Ib = 20 mAdc	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Emission and Power oscillation (1)	--- ---	--- ---	Is Po	80 14	--- ---	mAdc W

NOTES:

1. Ground the negative terminal of the dc filament.
2. No operating period or "on" period shall exceed 5 minutes, and every "on" period shall be followed by an "off" or standby period of at least the same, or greater, duration.
3. This test to be performed at the conclusion of the holding period.
4. Base sleeve (pin No. 8) tied to ground.
5. The hexaphase short and discontinuity test may be used as an alternate equivalent test for transmitting tubes. (See figure 2.)

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11,
DSA - ES

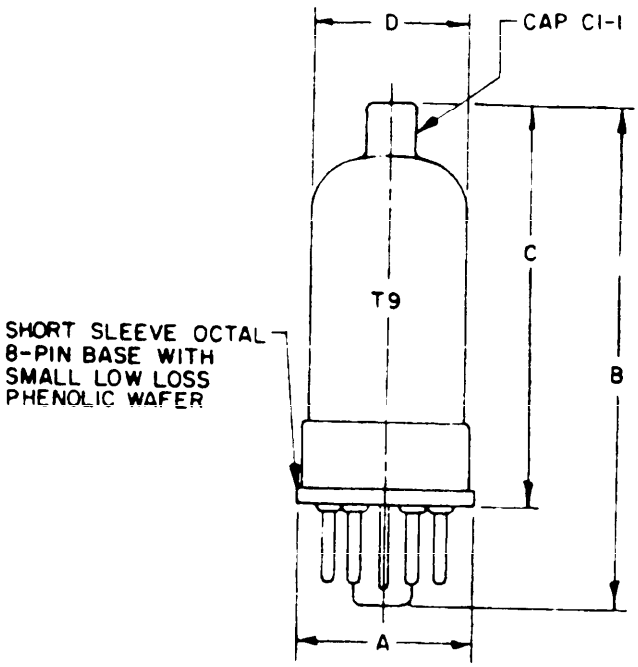
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-76)

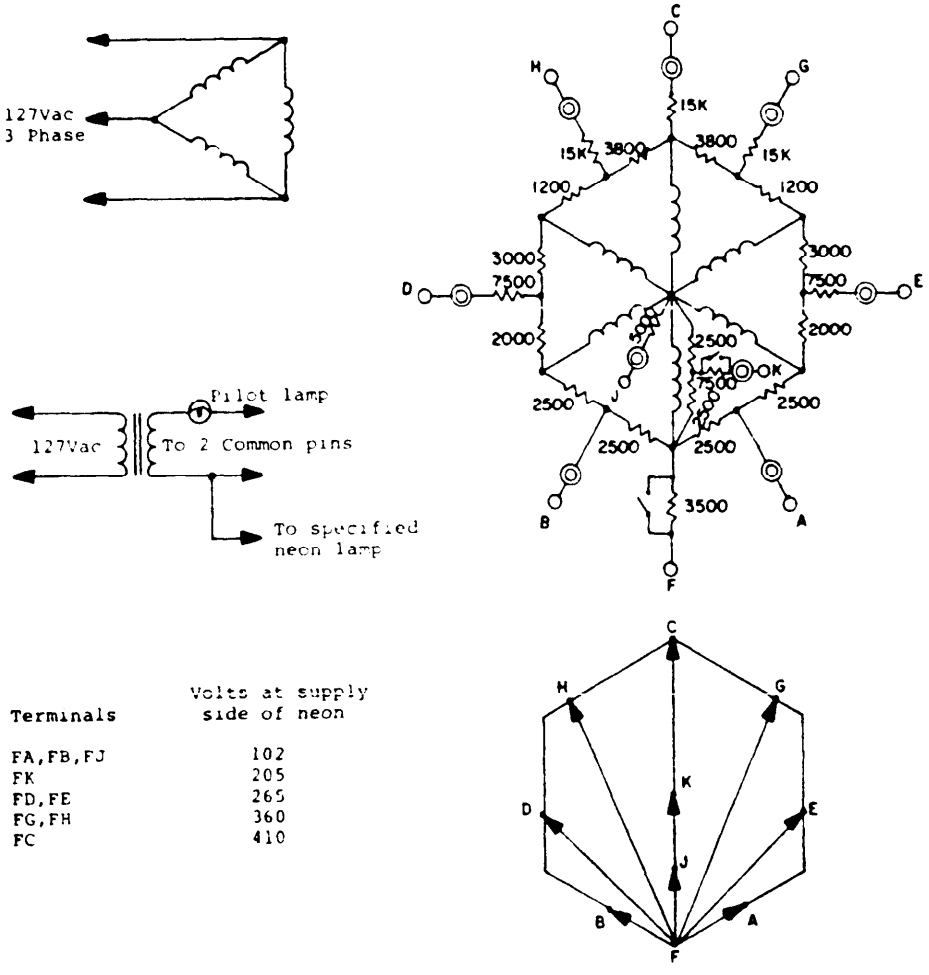
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
A		1.312 (33.32)
Quality conformance inspection, part 2		
B	3.344 (84.94)	3.656 (92.86)
C	2.781 (70.64)	3.094 (78.59)
D		1.188 (30.18)

FIGURE 1. Outline drawing of electron tube type 2E24.

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Terminals	Volts at supply side of neon
FA,FB,FJ	102
FK	205
FD,FE	265
FG,FH	360
FC	410

TEST CIRCUIT, VECTOR DIAGRAM, AND VOLTAGES

NOTES:
a. Voltage tolerance [illegible] per cent. Measure voltages without a tube in the socket, using a rectifier type voltmeter (1000 ohms per volt minimum) or an average-reading type VTVM, such as the Hewlett-Packard 400 series, Ballantine 300 or 310, or the RCA Junior Voltohmyst or equivalent. Peak reading meters such as the RCA Senior Voltohmyst will not agree with the above figures for voltage at the terminals. The Voltohmyst shall be isolated from ground and from the technician during the measurement, as its case is connected to its common test lead.
b. Use a hexaphase supply connected to a ring potentiometer. Each tube element (except cathode) is connected through [illegible] ohms to an appropriate point on the ring potentiometer.

FIGURE 2. Hexaphase short and discontinuity test.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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